



The
BRITISH
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IN EGYPT

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Measurements and Analysis

Distance using ultrasonic sensor

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Theoretical background

Distance is the measurement between any two points in the space. There are many ways to measure the distance between objects or points depending on how long objects are far from each other. A ruler can measure the distance for objects not so far, but for objects far from each other, a radar could take place to measure the distance. There are constants for metric measurement, distance is always positive, distance from any object to itself is always zero and the distance between point A to point B is equal to point B to point A. Distance could also be the amount of movement that the object covered while moving. Physics is using distance in many studies like mechanics and atomic structure.

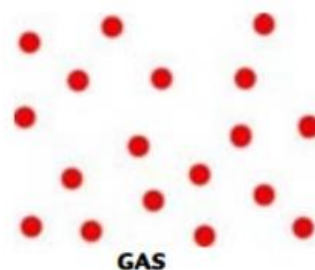
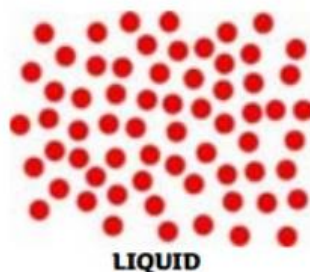
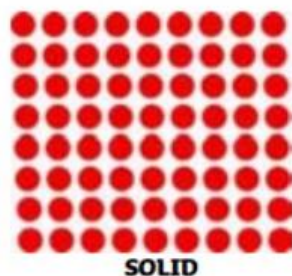
In Mechanics

the distance is used combined with time to study the motion of objects.

$$\text{distance} = \text{velocity} \times \text{time}$$

In atomic structure

Distance between atoms has a high effect on the structure of matter. The object could be solid, liquid, or gas by knowing the distances between the atoms of any matter.



distance tools

There are various types of tools to measure distance with different principles here are some:

- 1) Laser range finder
- 2) Tape measurement tools
- 3) Surveyor's wheel

working principle

1) Laser range finder

The device emits a laser beam to the targeted object and then reflects it back to a receiver, knowing the speed of light ($c = 3 \times 10^8$) and the time between emitting and receiving the light beam (T) and calculating the distance (D).



The equation of distance by laser range finder is:

$$D = \frac{CT}{2}$$

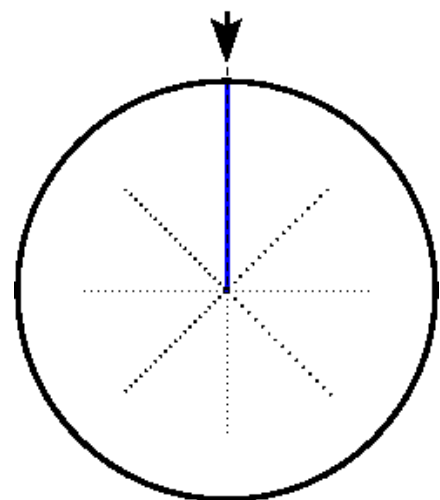
2) Tape measurement tools

Tape regularly made of metal or plastic printed on it repeated intervals of meters or centimeters as numbers. The tape is flexible and measures the length between two points by stretching the flexible strip between them.



3) Surveyor's wheel

Surveyor's wheel is a device with repeated intervals of radian angles on a wheel circumference. A counter is attached to the wheel and counts the intervals from the origin to the reached angle and the number of full rounds which is then presented as the distance on LCD screen.

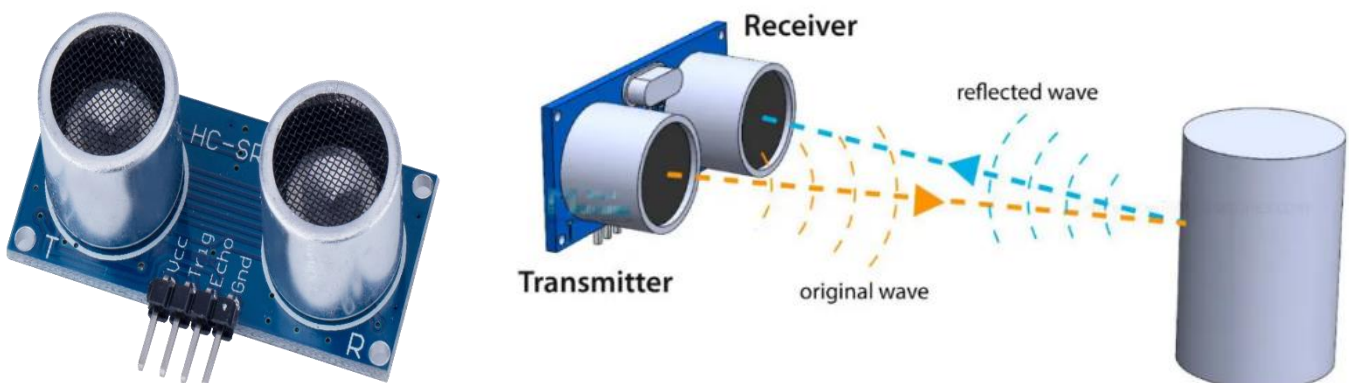


How to make measurement tools for distance?

Distance is a physical variable that represents numerically the space between two points or the orientation of a moving object through time. Tools measuring distance could measure space between two points directly by any type of repeated intervals of scale and add these intervals together and the summation will be the distance. There is another way by giving a reference plane for two points, by knowing their orientation from the reference and subtracting the second point orientation from the first point orientation the solution will be the distance. Another way to find the distance is by making a relation between position and time to find the distance from a certain origin. Making any distance measurement tool should include any of the previous rules.

Ultrasonic distance sensor

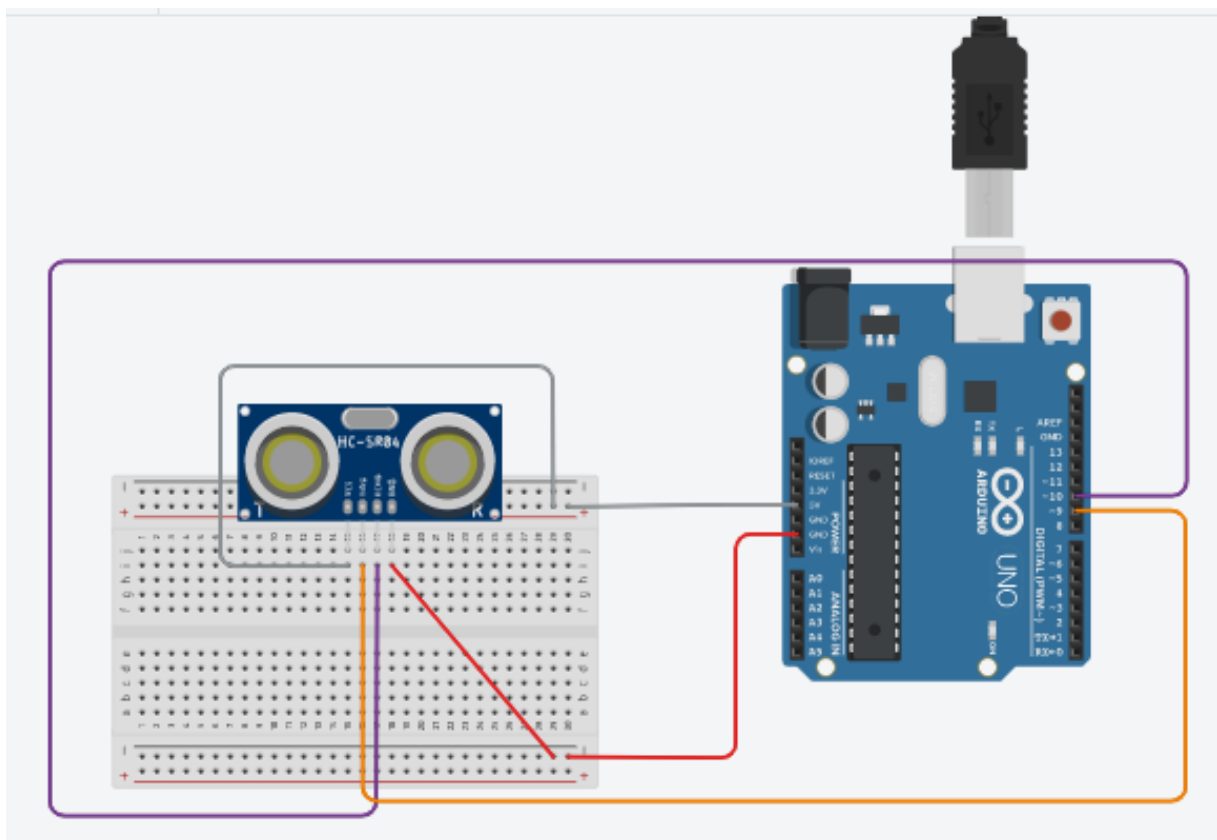
The ultrasonic distance sensor is a device that works by sending high-frequency sound waves. Humans cannot hear these waves because they are higher than the limit of human hearing (above 20 kilohertz). The ultrasonic sensor consists of four pins to be controlled (ground (GND), a voltage source (VCC), a trigger (Trig), and an echo receiver (Echo). The trigger and echo are the main components of the sensor. The trigger is a component that emits high-frequency sound waves until it reaches a targeted object. The sound waves are then reflected to the sensor as an echo reflection. The time taken in this process is calculated by the sensor finally the distance is determined by multiplying the speed of sound (about 343 m/s in the air) by the time taken divided by 2 this is because the distance is twice. The first time is when the sound wave transmits to the targeted object and the second is when it reflects going back to the sensor. The ultrasonic distance sensor measures the distance range from (2 cm to 400 cm) with an accuracy of (3 mm) and an operating frequency of (40 hertz).



Prototype circuit

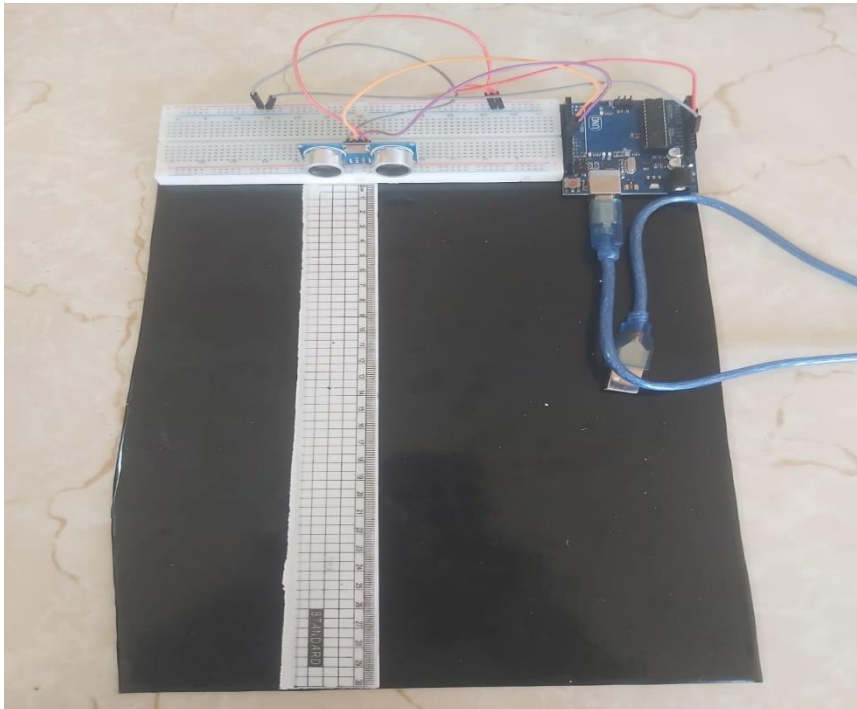
Components used

- 1) 2 grey jumper wires
- 2) 2 red jumper wires
- 3) 1 purple jumper wire
- 4) 1 orange jumper wire
- 5) Arduino Uno
- 6) Breadboard
- 7) Ultrasonic distance sensor

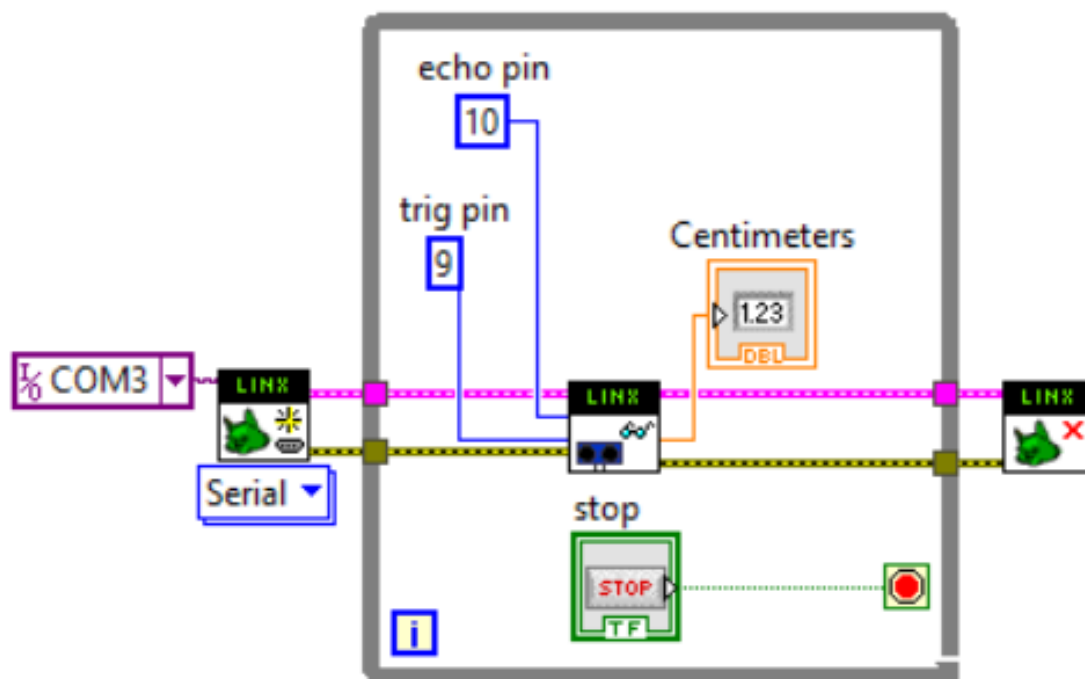


- 1) The two red jumper wires are used to attach the (GND) of the sensor to the (GND) of the Arduino at the common (-) node in the breadboard.
- 2) The two grey jumper wires are used to attach the (VCC) of the sensor to the (5V) of the Arduino at the common (+) node in the breadboard.
- 3) The orange jumper wire is used to attach the (Trig) of the sensor to the pin number (9) in the Arduino.
- 4) The purple jumper wire is used to attach the (Echo) of the sensor to the pin number (10) in the Arduino.

Real prototype circuit

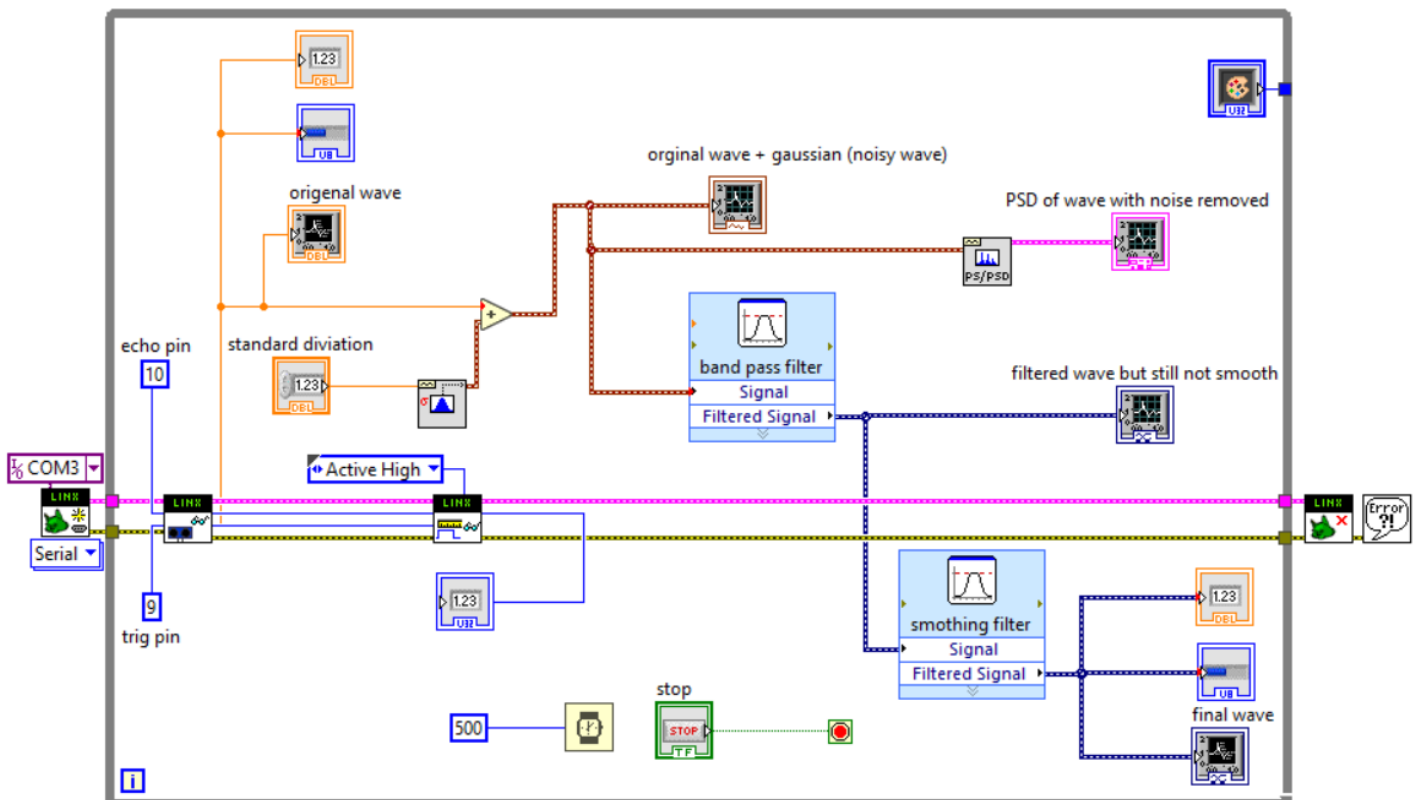


Data acquisition and interfacing with LabVIEW



The main block diagram for the ultrasonic distance sensor

Block diagram



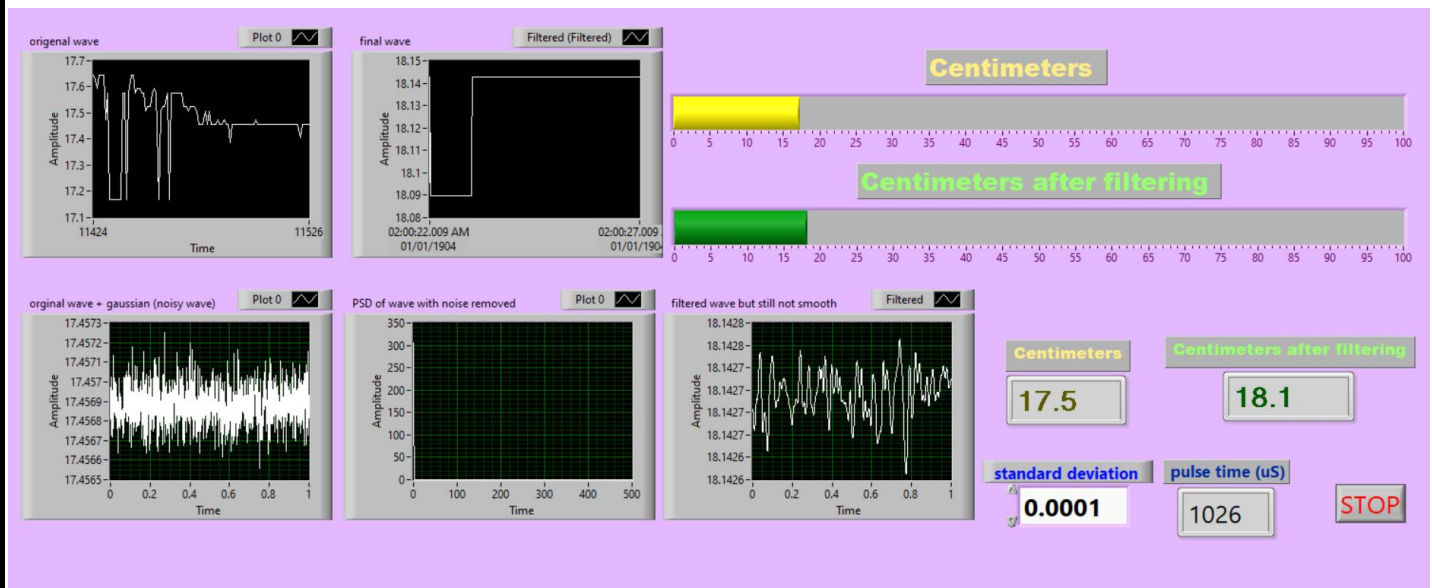
The block diagram consists of the main block diagram of the ultrasonic sensor and filters to filter the sound waves to take accurate reads for distance. Digital read pulse width is also added to show the timing of the total distance covered by the waves from the trigger until it is back as an echo to the sensor the distance could be calculated by the following equation.

$$\text{Distance} = \frac{\text{Time}}{2} \times 10^{-6} \times \text{speed of sound}$$

Filter circuit steps (based on PSD)

- 1) Gaussian wave is added to the original wave to cover the boundaries of the original wave because it is noisy and has a very high frequency, so it makes the PSD filter to be more effective in filtering and covers all the boundaries of the original wave. Standard deviation is added to be controlled to control the Gaussian wave width the more it is low the more accurate the readings of the distance this is because the Gaussian waves generated more and cover more boundaries.
- 2) PSD then takes place to remove the noise of the Gaussian waveform in addition to removing certain unneeded frequencies to make the wave ready to be processed and filtered.
- 3) Band pass filter is then added to let certain frequencies pass within a range and blocks the frequencies, not interested range, and blocks any remaining noises. FIR filter is part of the bandpass filter. The FIR filter makes the wave linear without any distortion and stable, but it does not cover long ranges.
- 4) Finally smoothing filter is added to smooth the wave by rippling the wave and covering any errors of long range that the FIR filter didn't cover properly.

Front panel



The front panel consists of two waveform charts representing the original wave and the final wave after filtering for measuring distance in centimetres. three waveform graphs are also added to illustrate the waves during the filtering processes. Three indicators are added to print the distances. The yellow indicator is for the original distance from the sensor and the green is for the distance after the filtering processes the dark blue indicator illustrates the trip of the wave from the trigger to the echo time taken. Control is added for the standard deviation to condition the Gaussian wave in case of any inaccuracy greater than 3mm. 100 cm horizontal bars are added to give a better vision view. A colour box is added for the background. Finally, a stop button is added to stop the program.

Conclusion

In conclusion, Distance is a physical quantity represented numerically to show how far any object or point is covered apart. The distance can also be how far any two objects far from each other. LabVIEW is a tool used combined with the Linx toolkit to illustrate physical quantities. An ultrasonic sensor was used to illustrate the distance instead of regular tools. Arduino was used to make a circuit and interfaced with LabVIEW.

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